Topic B2: Scaling up

B2.1 Supplying the cell

Summary

Cells transport many substances across their membranes by diffusion, osmosis and active transport. Stem cells are found in both plants and animals. These stem cells can divide, differentiate and become specialised to form tissues, organs and organ systems.

Underlying knowledge and understanding

Learners should be familiar with the role of diffusion in the movement of materials in and between cells.

Common misconceptions

Learners commonly show some confusion regarding surface area: volume ratio, particularly how larger animals have a smaller surface area: volume ratio. They also show some confusion as to stem cells: where they are found and their roles. Care should be taken to give clear definitions when covering this content.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM2.1i	use percentiles and calculate percentage gain and loss of mass	M1c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
B2.1a	explain how substances are transported into and out of cells through diffusion, osmosis and active transport	examples of substances moved, direction of movement, concentration gradients and use of the term water potential (no mathematical use of water potential required)	M1c, M1d	WS2a, WS2b, WS2c, WS2d	Observation of osmosis in plant cells using a light microscope. Investigation of 'creaming yeast' to show osmosis. (PAG B6, PAG B8) Investigation into changes in mass of vegetable chips when placed in sucrose/salt concentrations of varying concentrations. (PAG B6, PAG B8)
B2.1b	describe the process of mitosis in growth, including the cell cycle	the stages of the cell cycle as cell growth, DNA replication, more cell growth, movement of chromosomes		WS2a, WS2b, WS2c, WS2d	Modelling of mitosis using everyday objects e.g. shoes, socks etc. Observation of mitosis in stained root tip cells. (PAG B1, PAG B6, PAG B7)
B2.1c	explain the importance of cell differentiation	the production of specialised cells allowing organisms to become more efficient and examples of specialised cells		WS2a, WS2b, WS2c, WS2d	Examination of a range of specialised cells using a light microscope. (PAG B1)
B2.1d	recall that stem cells are present in embryonic and adult animals, and meristems in plants				Demonstration of cloning using cauliflower. (PAG B6, PAG B7)
B2.1e	describe the functions of stem cells in embryonic and adult animals, and meristems in plants	division to produce a range of different cell types for development, growth and repair		WS1.1e, WS1.1f, WS1.1h	
B2.1f	describe the difference between embryonic and adult stem cells in animals				Research into the different types of stem cells.